

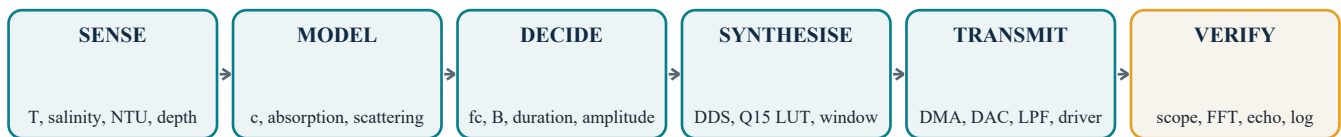
1 The problem

- Fixed-frequency sonar cannot preserve both range and resolution as temperature, salinity, depth, and turbidity change.
- SeaNergy measures those conditions and solves for centre frequency, bandwidth, duration, amplitude, and window before each ping.
- The payload targets real-time output using timers, DMA, a DAC, filtering, amplification, and a water-coupled transducer.

2 What we built

The bench stack reads temperature, TDS, turbidity, and three analog control inputs on an ESP32-S3. The application executes ocean physics, waveform selection, synthesis, matched filtering, power estimation, and closed-loop correction offline.

3 System architecture

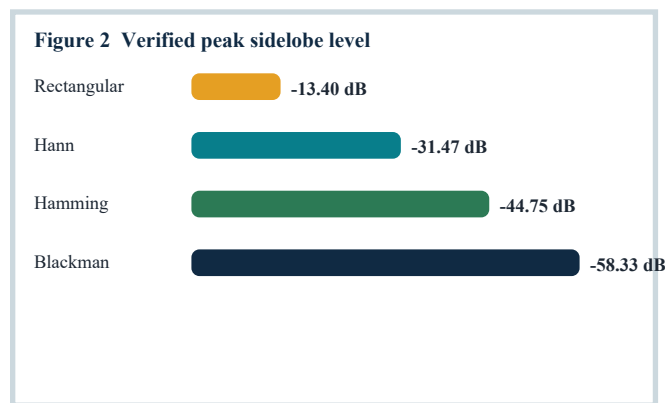
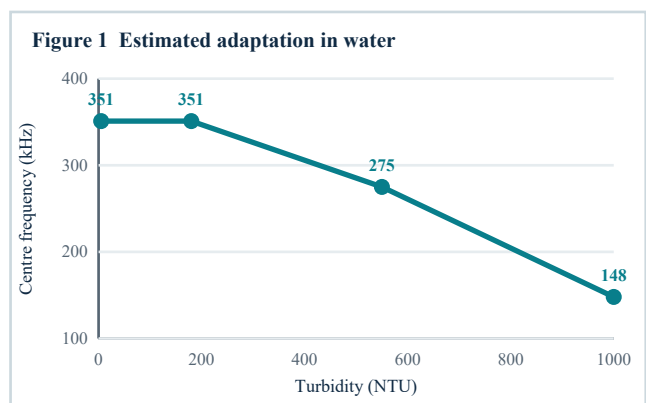


4 Capability now

- 85 physics and core checks pass with zero failures.
- 33 screen cases pass across 12 application screens.
- CW, LFM, geometric, and Barker-13 modes run in software.
- The solver evaluates about 470 parameter sets per decision.
- The sensor and analog-control bench path is operating.
- The V1 enclosure geometry has passed inspection.

5 Evidence and estimated achievement

Metric	Result	Provenance
Turbidity response	351 -> 148 kHz from 5 -> 1,000 NTU	Modelled water sweep
Design compression	TBP 240.5; gain 23.81 dB	Calculated
Design resolution	1.08 cm; about 248x over uncompressed	Calculated
Window sidelobes	-13.40 / -31.47 / -44.75 / -58.33 dB	Verified software
Barker-13 PSLR	22.28 dB	Verified software
Power and endurance	5.58 mJ/ping; 7.26 h at 10 s	Modelled
Verification	85 core + 33 screen cases; zero failures	Executed
Cost	INR 2,500-4,500 bench; INR 36,000 marine	Planning estimate



6 Reality ledger

Real today	Estimated or open
Sensor bench path; model; app; four waveform families; automated checks; V1 geometry.	Water adaptation; absolute power; DDS/DMA hardware stream; amplitude control; THD; SFDR; pressure sealing.

7 Path to deployment

1. Land the DDS/DMA transmitter and complete TX-001 to TX-004 on a scope.
2. Fit a water-coupled 200 kHz transducer pair and complete the controlled tank campaign.
3. Build the sealed V2 pod with current sensing, wet interfaces, and pressure qualification.